Platform User Guide

Recording, scoring and reviewing simulated suturing practice

For students and instructors · October 2026



What SutureAI does. Students record a practice task (simple suturing, knot tying or needle handling) and upload the video. The platform scores it on five dimensions and writes a report that links each finding to a moment in the video. Instructors follow their whole class, confirm or adjust scores, and decide students' appeals against individual findings.

Students	Section 03: dashboard, uploading a recording, reading a report, appealing a finding, exporting a PDF.
Instructors	Section 04: class overview, student detail, review queue, deciding appeals, reviewing a report.
Everyone	Sections 02 and 05: signing in, statuses, troubleshooting.

01 How it works

Every recording goes through the same loop. The student only does two things (upload, then respond); scoring and the first draft of the report are automatic; the instructor has the final word on the score and decides any appeal against a finding.

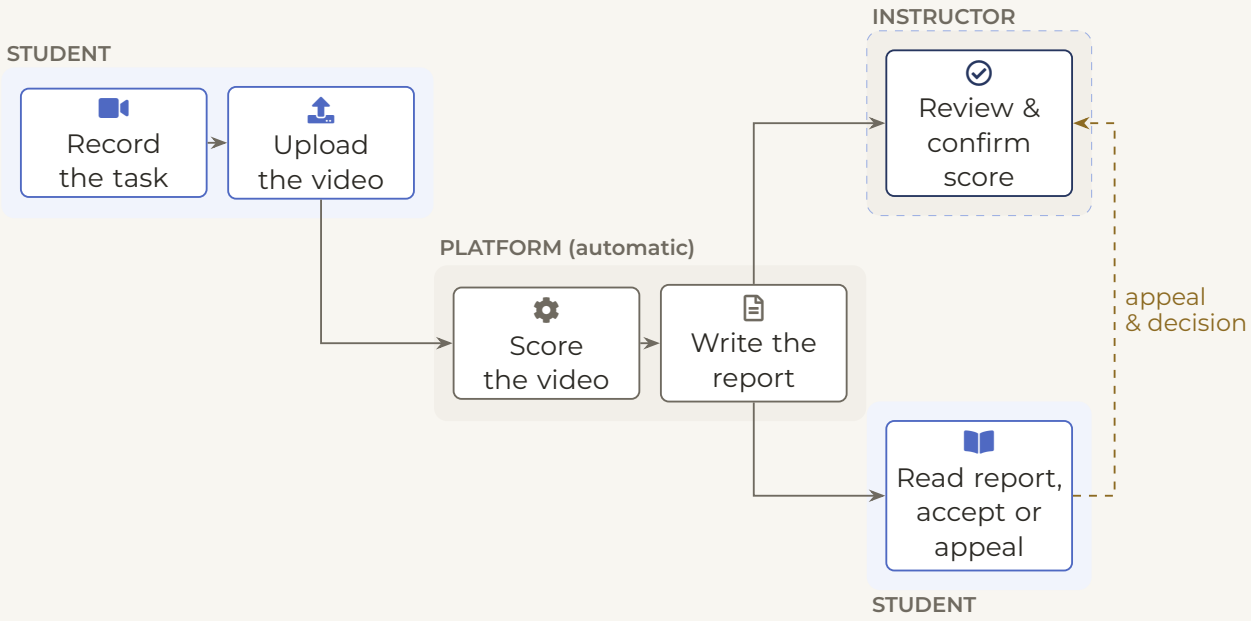


Figure 1: The training loop. Scoring and report writing run in the background; the student can close the page and come back.

The report scores five **dimensions**, each from 0 to 100:

Efficiency	Duration, path length of the instruments, idle time between steps.
Accuracy	Needle entry and exit points, repeated insertion attempts, stitch spacing.
Procedural Flow	Whether every step was done, in the right order, without rework.
Safety	Dropped needle, wrong instrument, instruments leaving the working area.
Product Quality	Consistency of the finished stitches and security of the knots.

Prototype notice

In this version the video scoring step is a placeholder: scores and detected events come from a simulation (or, for the demonstration recording, from a manual annotation) rather than from analysing the video. Everything else (upload, reports, appeals, reviews, PDF export) works as described.

02 Getting started

02.1 Signing in

Open the site and enter your email and password on the sign-in card, then press **Sign in**. You land on the page for your role: students on their *dashboard*, instructors on the *class overview*. The **中文 / EN** switch in the corner of the card (and at the top of every page) changes the interface language at any time.

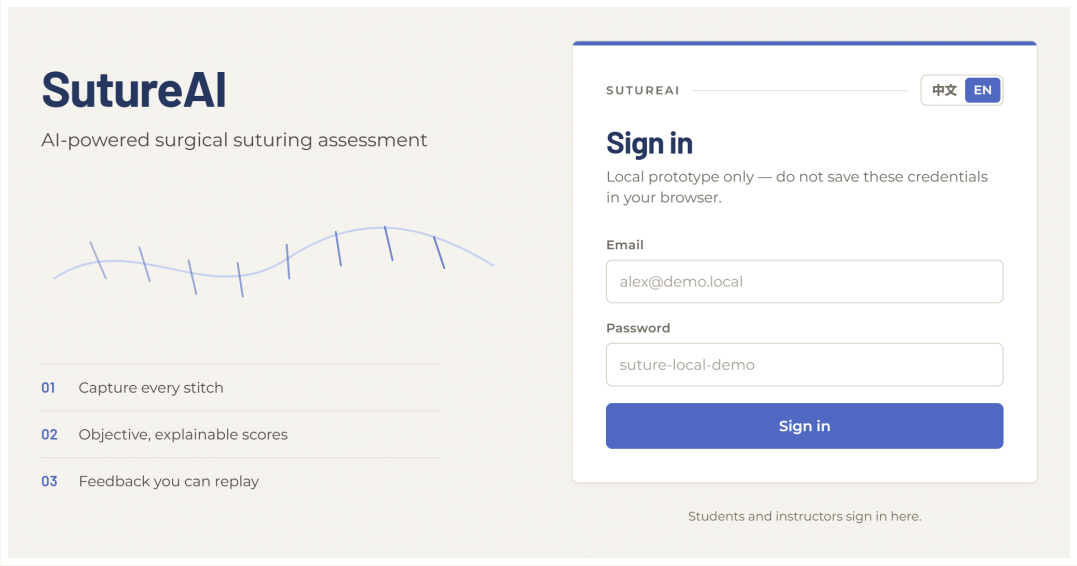


Figure 2: Sign-in page.

The demo installation comes with these accounts (password `suture-local-demo`):

Email	Who
`chen@demo.local`	Instructor, Groups A and B
`patel@demo.local`	Instructor, Group B only
`alex@demo.local`	Student, Group A
`priya@demo.local`	Student, scores declining
`sam@demo.local`	Student, improving
`jordan@demo.local`	Student, Group B

Good to know

Instructors only ever see students in the groups they teach.

To leave, press **Log out** in the top bar.

03 For students

03.1 Your dashboard

The dashboard (/) is your home page. A slim top bar holds the language switch, your name and **Log out**. Below it, the welcome sheet shows your course and training focus next to a dashed *Current training task* tile with the two main actions: **Start training** records a new session, **Open latest report** jumps to your most recent report. If a recording is still being processed, a *Recordings in progress* list appears here too.

The numbered sections below it are: 01 *Task details* (task, status, due date, estimated duration and the dimensions it is scored on), 02 *Latest report snapshot*, 03 *Recent training trend*, 04 *Next practice priorities* taken from your last report, 05 *Instructor & system notices* (for example when your instructor has reviewed a report or decided one of your appeals), 06 *Training activity calendar* and 07 *Training history* with every report.

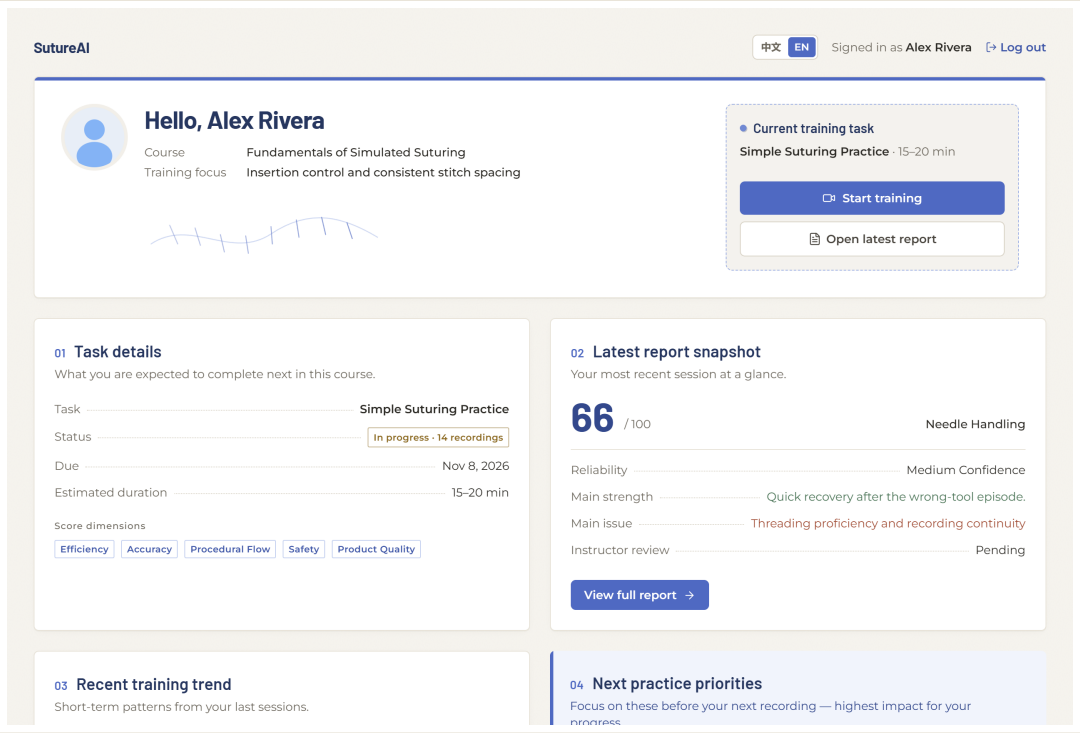


Figure 3: Student dashboard. The dashed tile on the right holds **Start training** and **Open latest report**.

03.2 Recording and uploading a session

1. On the dashboard press **Start training** in the *Current training task* tile.
2. Under 01 *Task*, pick the task you practised. The current task is already selected.
3. Under 02 *Recording*, add your video: drag it onto the dashed area or click it to browse. MP4, WebM, MOV and MKV are accepted. A preview appears with the file name, size and duration so you can check it is the right recording.
4. Press **Upload & analyse**. A progress bar shows the upload; you can cancel it.

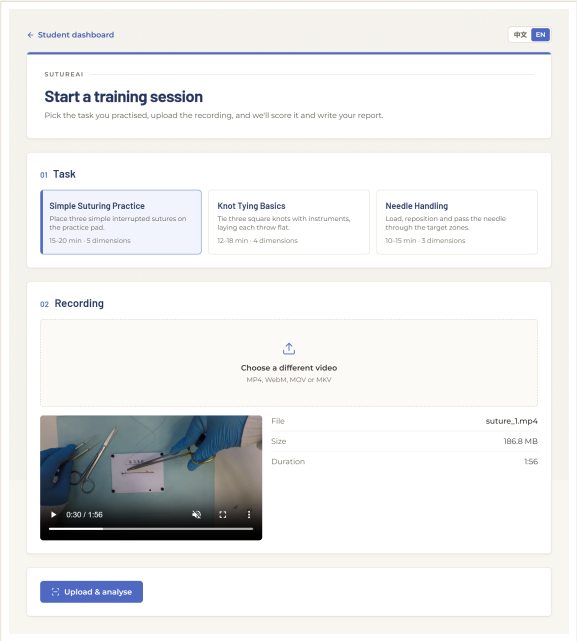


Figure 4: Choosing the task and the video before uploading.

After the upload the page shows the analysis as it happens: a large percentage with an estimate of the time left, a progress bar drawn as a line of stitches, a checklist of the seven analysis stages (verify the recording, detect instruments, track hands, segment the procedure steps, score the dimensions, write the report, link evidence clips) and a dark *Activity* panel describing what is being done right now. It usually takes under a minute. When it finishes, the report opens automatically.

You can leave the page at any time: unfinished sessions are listed under *Recordings in progress* on your dashboard, and the page address (`/train/<session>`) can be reopened later.

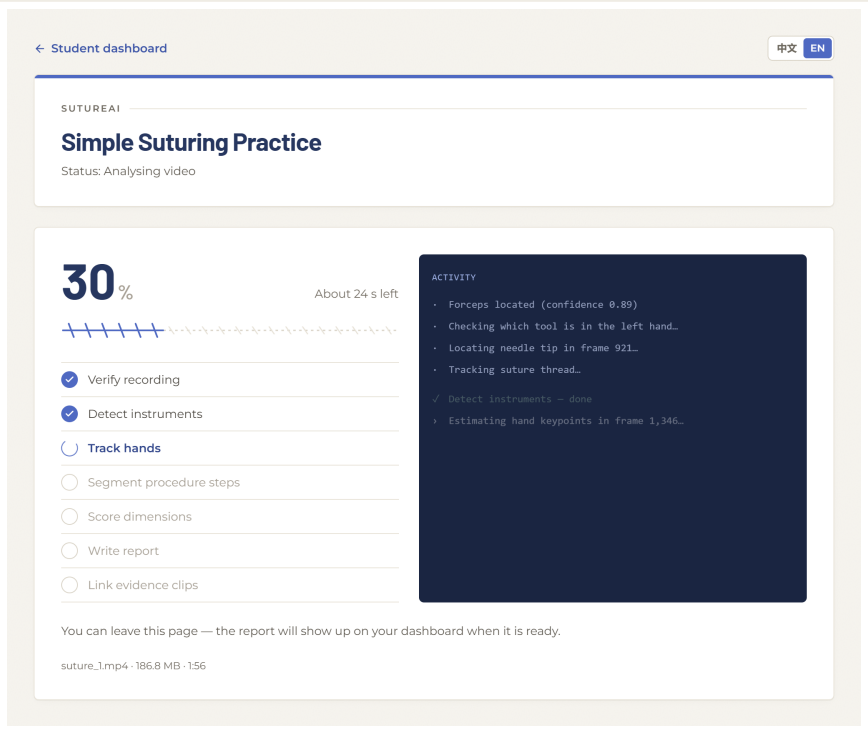


Figure 5: Analysis in progress: percentage, stitch progress bar and stage checklist on the left, live activity on the right.

> **Tip**
> For the most reliable scores, film from above with the whole practice pad in frame, keep your hands from covering the needle, use even lighting, and record the task from start to finish without stopping.

03.3 Reading your report

The header sheet of the report names the task, the student, the date and time, the task duration and how confident the system is. The rotated stamp in the top-right corner shows whether your instructor has looked at it yet: *Pending review* or *Reviewed* (hover over it for the full wording). Use the **Session** menu to switch between your reports and **Export PDF** to save one (Section 03.5).

Some reports then open with a *Fix this first* tile: the single change that would gain you the most points, why it matters, a drill to practise it and the moment it was seen. The numbered sections follow.

01 *Overall performance*: the overall score drawn as a ring, with reliability, change since your previous session, duration and dimension average. If your instructor confirmed a different score, or accepted one of your appeals, a blue *Current score* line shows the score that counts. A *Confidence notice* next to it lists anything that may limit the accuracy of the scores. Where available, *Where the points went* lists the biggest losses first, next to your progress since the last session.

02 *Dimension scores*: a radar of the five dimensions and one card per dimension; **Metrics** shows the measurements behind a score.

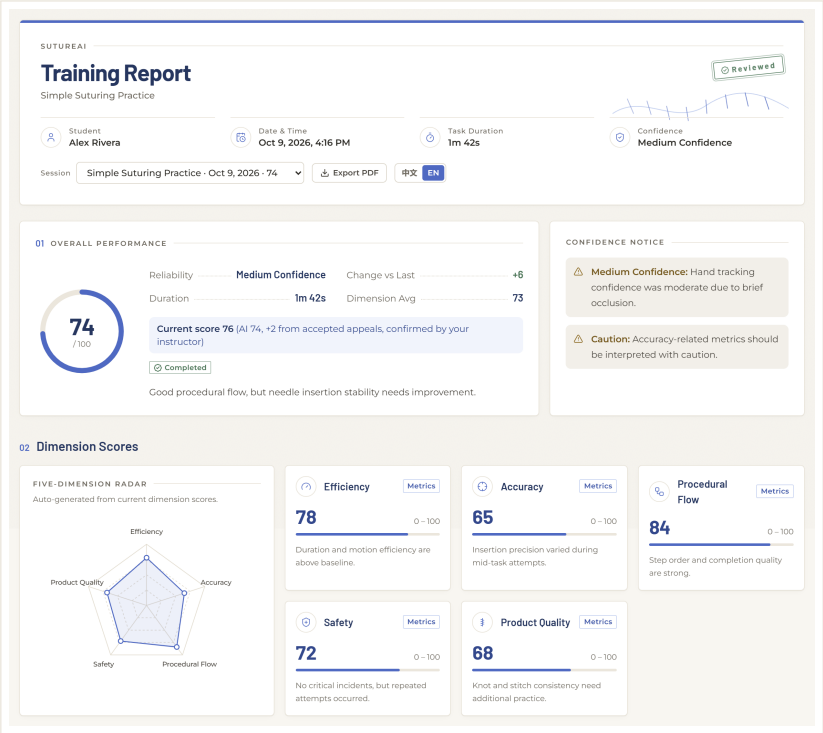


Figure 6: Report header with the *Reviewed* stamp, overall performance and dimension scores.

03 *Training timeline* lists what was detected and when: normal steps (hollow dot, *Step*), warnings (ochre, *Warning*) and issues (brick red, *Issue*). By default only warnings and issues are shown; **Show all *n* events** adds the normal steps. **Expand** shows the detail; **Replay** (or **Jump**) plays the recording from that moment; on a report without a recording the button reads **View Clip**. Warnings and issues also have an **Appeal** link (Section 03.4). *Assessment highlights* beside it summarise the strongest dimension, the main issue, what to focus on and the review status.

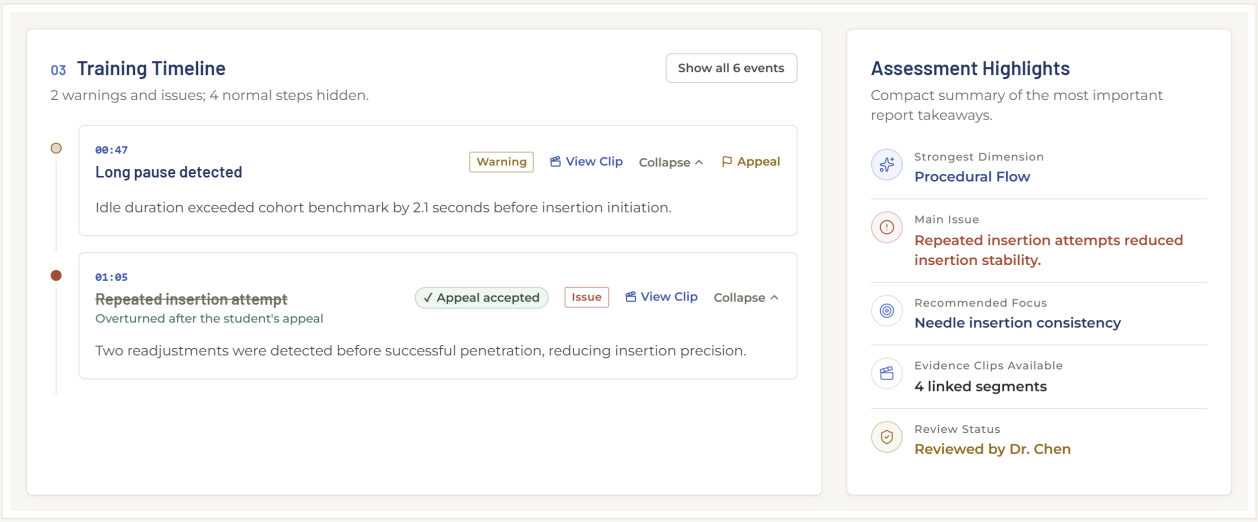


Figure 7: Timeline. The struck-through issue at 01:05 was overturned after the student's appeal; the warning at 00:47 can still be appealed.

04 *Evidence replay* holds your recording. The marker strip under the video shows every event (issues, warnings, steps); click a marker to jump there. Each key clip card's **Replay**

button plays the recording from that moment.

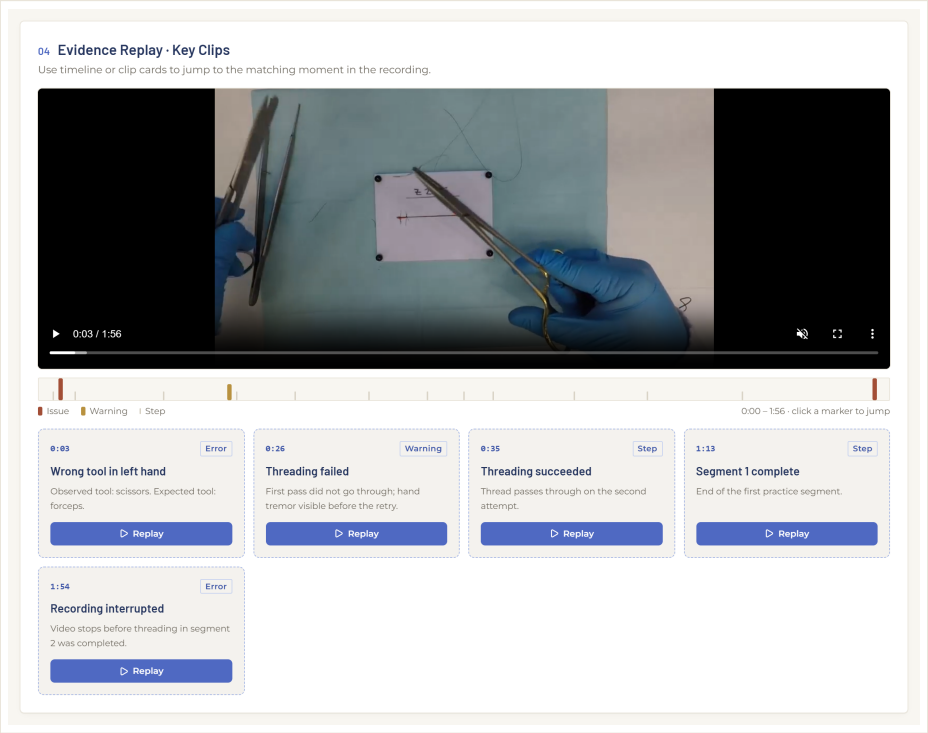


Figure 8: Evidence replay with the event marker strip and the key clips.

05 *Report feedback & reflection* closes the report. On the left are the AI feedback (blue tile), your strengths and areas for improvement (grey tiles) and recommended next steps (dashed tile); on the right are your instructor's evaluation, your response and your personal note (next section).

03.4 Accepting or appealing your report

If you agree with the report, press **Accept report** in the *Your response* tile of section 05.

If you think a finding is wrong, **appeal it**:

1. Find the warning or issue in the timeline (section 03) and press **Replay** to check the moment yourself.
2. Press **Appeal** on that event and choose a reason: *This didn't happen*, *The view was blocked*, *Intentional / wrong moment* or *Other*.
3. Explain what really happened and press **Send appeal**.

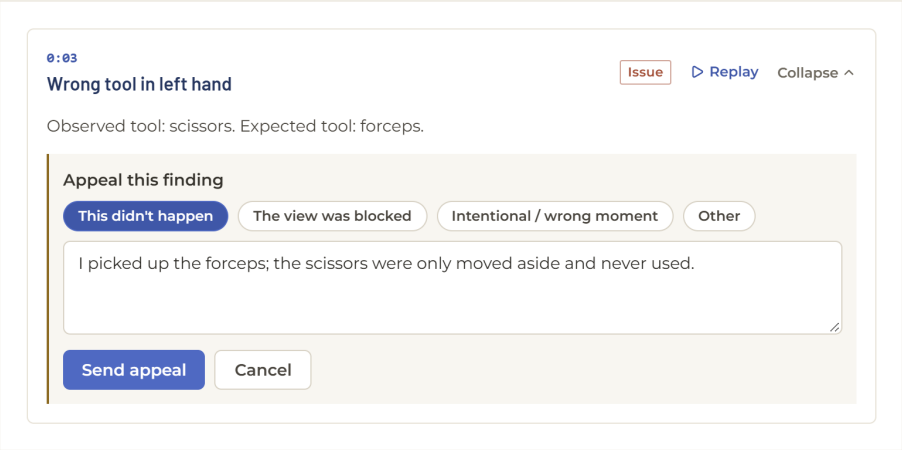


Figure 9: Appealing a finding directly from the timeline.

The event then shows Appeal pending and your instructor is notified. They watch the clip and decide:

- **Accepted**: the finding is struck through as *Overturned after the student's appeal* and your score is raised (the chip shows by how much, e.g. Appeal accepted · +2).
- **Rejected**: the finding stays; your instructor explains why.

Either way the decision and the instructor's reasoning appear under *Your response*. You can withdraw an appeal while it is still pending, and you can also appeal the report as a whole with **Appeal the whole report**. A report can carry up to five appeals.

Above it, **Instructor evaluation** shows your instructor's final score, comment and teaching focus once the review is complete. Below it, **Personal note** (dashed tile) is a private reflection saved with the report with **Save note**; instructors cannot see it.

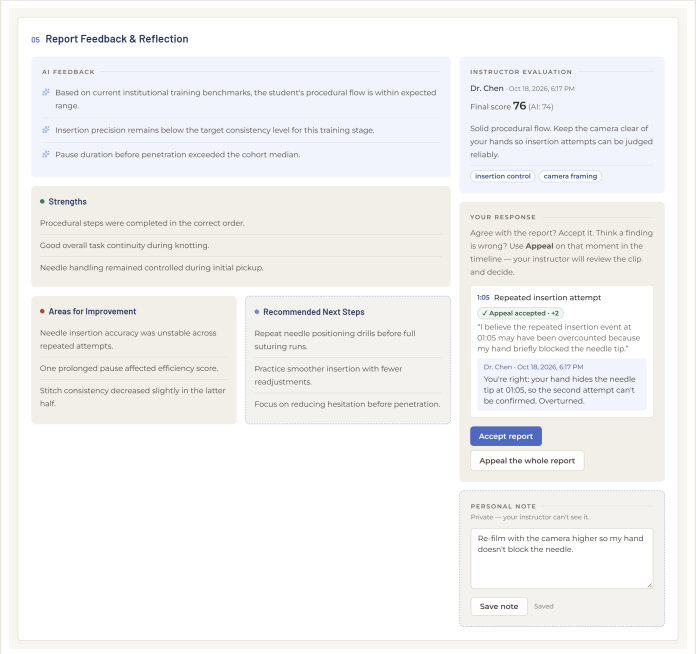


Figure 10: Section 05 of a reviewed report: the appeal on 01:05 was accepted (+2) with the instructor's reasoning, and the student kept a private note.

03.5 Exporting a report as PDF

Press **Export PDF** in the report header. A print layout opens in a new tab and the browser's print dialog appears; choose **Save as PDF** as the destination. The PDF contains the summary, what to fix first and where the points went, the dimension scores, the timeline with a still from the recording at every event, strengths and next steps, all appeals with their decisions and the instructor's review, so it can be filed as a record of the session.

The print page waits until the video stills have loaded. If the print dialog does not appear, press **Save as PDF** at the top of the page.


SUTUREAI · TRAINING REPORT

Simple Suturing Practice

Alex Rivera · Oct 18, 2026, 6:17 PM · 1m 56s (2 segments, second interrupted)

66
AI score
Pending Instructor Review

SUMMARY

Needle handling is still developing: noticeable tremor and uneven motion during threading. A brief wrong-tool pick was corrected quickly. The second segment ends with a recording interruption before threading was completed.

Confidence: Medium Confidence · Change vs last session: +0 · Task incomplete

FIX THIS FIRST

Threading needed more than one attempt · about −9 pts · at 0:26
Each failed pass adds time and shows unstable hand control near the needle eye.
Practice: Rest both wrists on the table and bring the thread to the needle, not the needle to the thread. Do 10 reps.

WHERE THE 34 POINTS WENT

Threading needed more than one attempt	−9	0:26
Recording stopped before the task was finished	−6	1:54
Picked up the wrong instrument	−5	0:03
Other technique gaps	−14	

× Idle time 21% (target ≤ 12%) × Threading attempts 2 (target 1) × Wrong-tool pickups 1 (target 0) × Rework 2 (target 0)

DIMENSION SCORES

Efficiency	62	Pauses and the repeated threading attempt slow the pace versus cohort norms.
Accuracy	64	Threading succeeded on the second attempt; micro-adjustments were unstable.
Procedural Flow	68	Knot sequence completed, but the initial tool selection was wrong.
Safety	72	No high-risk behaviour; the tool mix-up was corrected quickly.
Product Quality	64	Knot quality acceptable; segment 2 could not be assessed.

TIMELINE

0:02 Task start detected — Step
Segment 1 begins; both hands enter the working area.

0:03 Wrong tool in left hand — Issue
Observed tool: scissors. Expected tool: forceps.

0:05 Switched to correct tool — Step
Forceps picked up; task continues.

Figure 11: Start of an exported report. Each timeline event shows the video frame at that moment.

04 For instructors

The instructor area has two tabs in the top bar, **Overview** and **Review queue**; the language switch, your name and **Log out** sit on the right of the same bar. Both pages lead to student pages and reports:

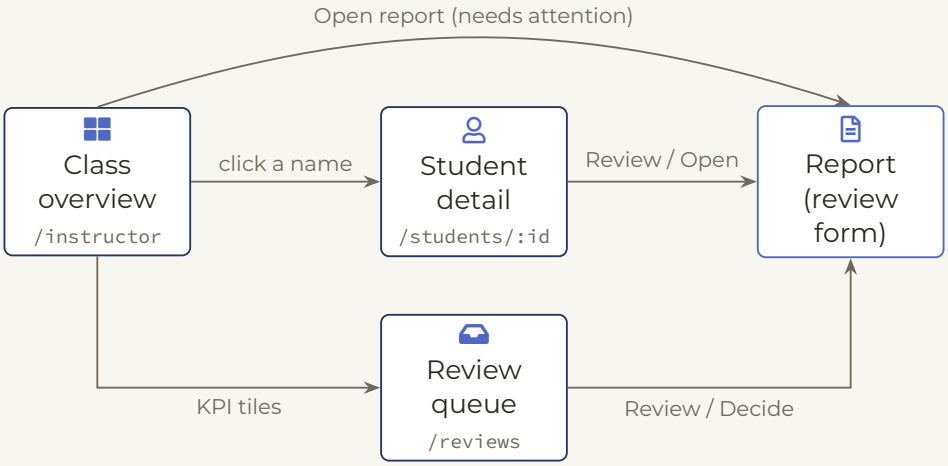


Figure 12: Navigating the instructor area.

04.1 Class overview

Choose a group with the **Cohort** menu at the top right; everything on the page refers to that group.

- **Key numbers**: students (and how many trained this week), sessions this week, the 30-day average score, reports waiting for your review and open appeals. The last two are links into the review queue.
- **Needs attention** lists the students to look at now, most urgent first, each with a severity chip and a link to the student or the report.
- **Skill map** shows every student's recent average in each dimension on a blue scale. Paler cells are lower scores, so weak spots (for one student, or for the whole group in a column) stand out.

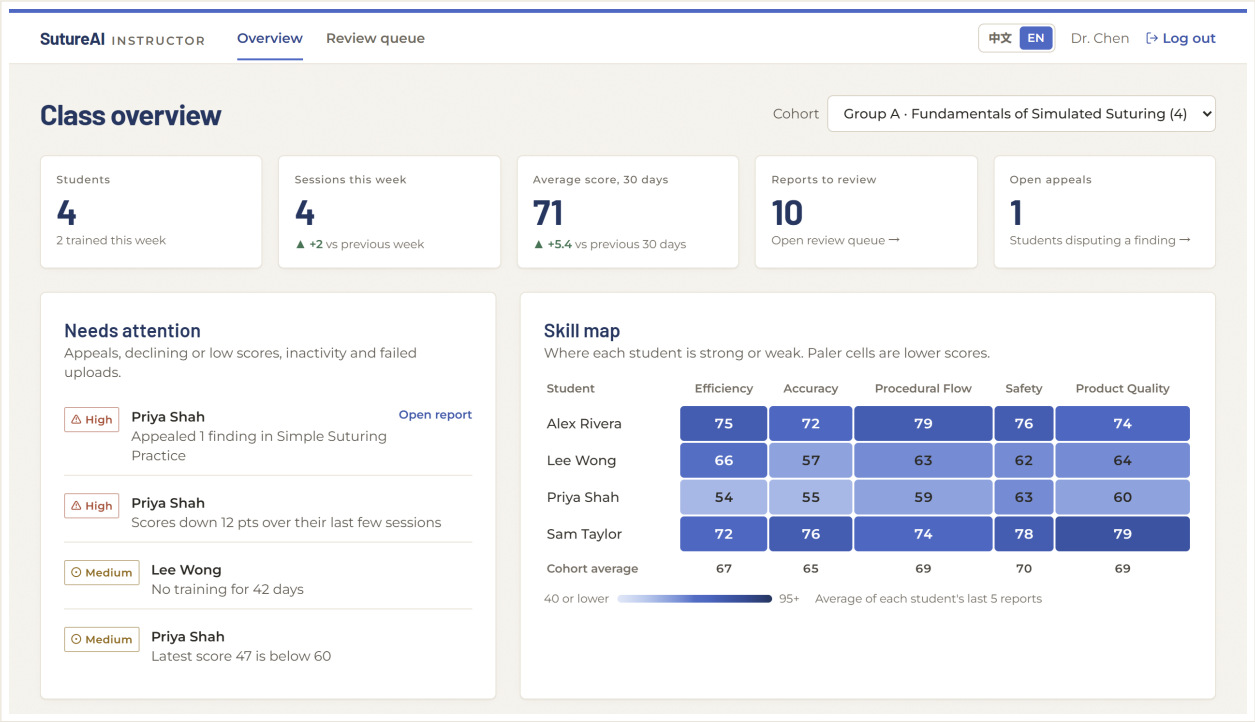


Figure 13: Class overview for Group A, below the instructor top bar.

Flag	Level	When it appears
Open appeal	High	The student appealed a finding and is waiting for your decision.
Declining scores	High	The average of the last 3 reports is at least 8 points below the 3 before them.
Low score	Medium	The latest score is below 60.
Inactive	Medium	No training for 14 days or more, or never.
Failed upload	Low	A recording could not be processed.

Below the skill map, the **Students** table lists everyone in the group. Click a column heading to sort (by default the students with the most waiting work come first) or type in the search box. *Trend* is a small line of the last eight scores; *Change* compares the average of the last three reports with the three before them; *To review* shows how many reports and appeals are waiting for you.

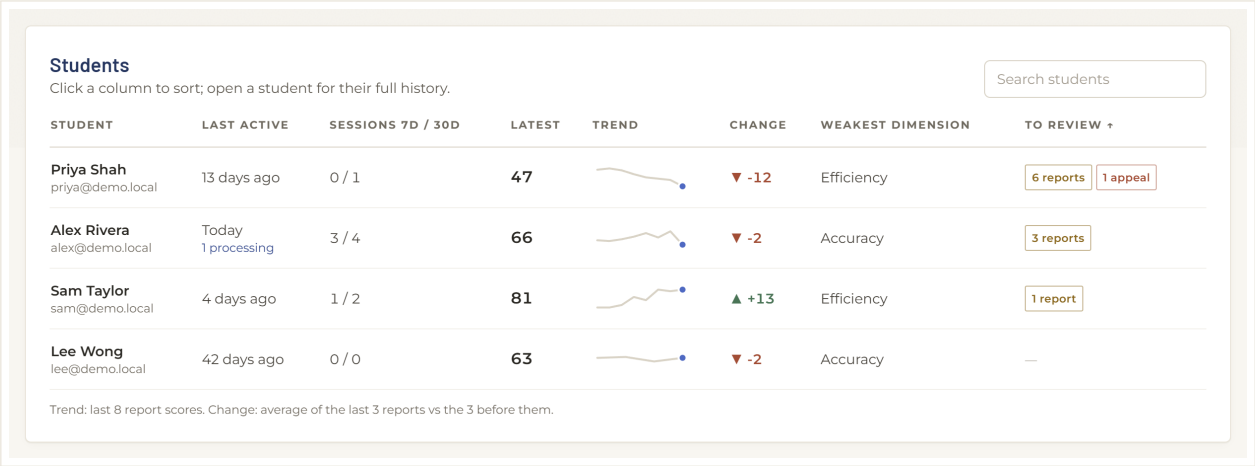


Figure 14: Student roster.

04.2 Student detail

Click a student's name anywhere to open their page: who they are, their group, current task and focus, and their current flags; key numbers (latest score, reports, weakest dimension and what is waiting on you); the overall score for every report (hover for details, click a point to open that report); one small chart per dimension to see *where* a change comes from; every session with its review and response status; and the training calendar.

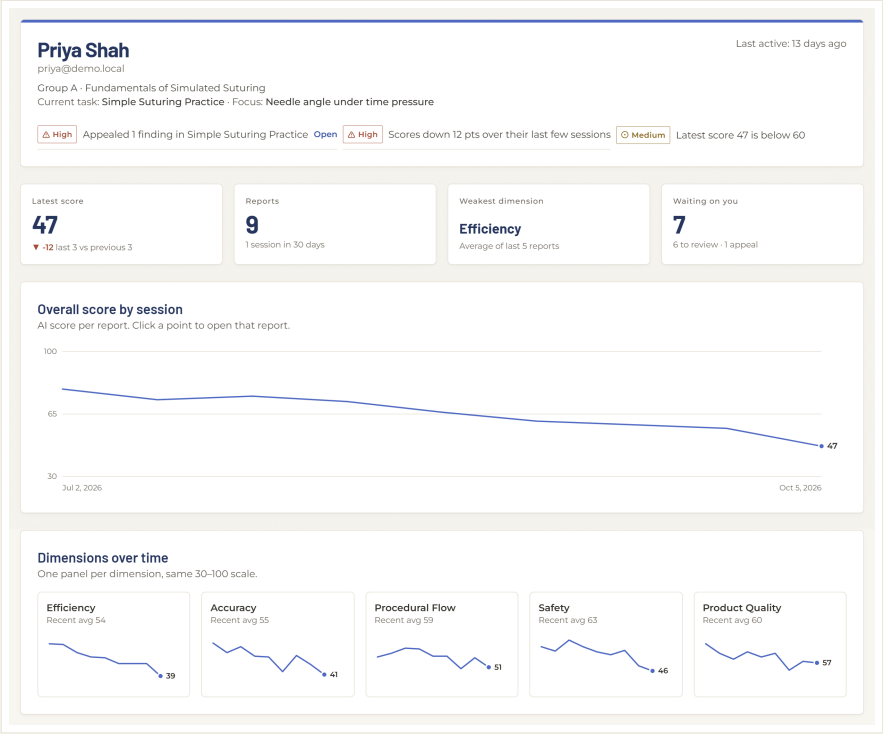


Figure 15: A student whose scores are declining. The dimension charts show the drop is broad rather than in one skill.

04.3 Review queue

The queue collects reports from all your groups (or one, with the **Cohort** filter) in three tabs, each with its count:

- **To review**: reports without a completed review, oldest first. Drafts stay here until completed.
- **Appeals**: reports with at least one appeal waiting for your decision.
- **Reviewed**: completed reviews, most recent first.

Each row ends with the action for that tab: **Review**, **Decide** or **Open**.

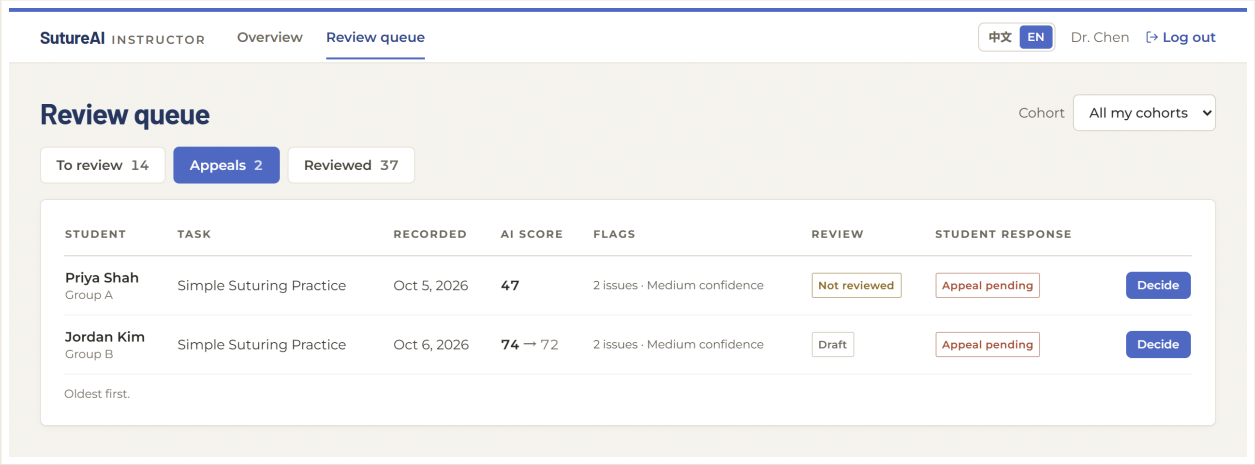


Figure 16: The *Appeals* tab of the review queue. Arrows such as *74 → 72* show a score you changed in a draft.

04.4 Reviewing a report

Opening a report as an instructor turns its last section into 05 *Report feedback & review*: the AI feedback on the left, *Student appeals* on the right and the *Instructor review* form below.

1. **Decide any appeals** under *Student appeals*. Each one shows the disputed finding, the student's reason and message; when the report has a recording, the time button (e.g. ▶ **0:03**) replays that moment. Choose **Accept — overturn this finding** (the score is adjusted by the amount you set; a value is suggested: +4 for an issue, +2 for a warning) or **Reject — keep the finding**, explain your decision, and press **Send decision**.
2. Set the **final score**. It starts at the AI score plus any accepted appeals; the *Alignment* tile updates as you type.
3. Write a **comment to the student** and add one or more **teaching focus** tags: type one and press Enter, or pick one of the suggested dimensions under the field.
4. Press **Save draft** to come back later (the student cannot see drafts), or **Complete review** to send it. A completed review can still be edited with **Update review**.

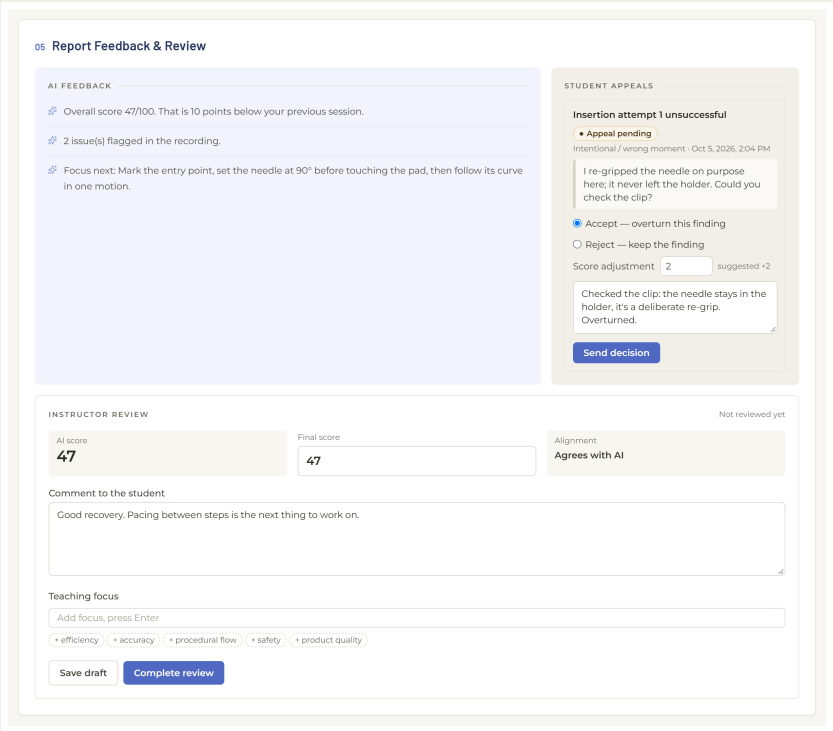


Figure 17: An appeal waiting for a decision, next to the AI feedback and above the review form.

Alignment	Meaning
Agrees with AI	Final score equals the AI score.
Adjusted (≤10 pts)	Changed by 10 points or less.
Override (>10 pts)	Changed by more than 10 points.

If you accept an appeal after completing the review, the final score is adjusted automatically.

- **Tip**

The View switch in the report header toggles between Instructor and As student sees it, which previews exactly what the student will see. Handy before completing a review.

05 Reference

05.1 Session status

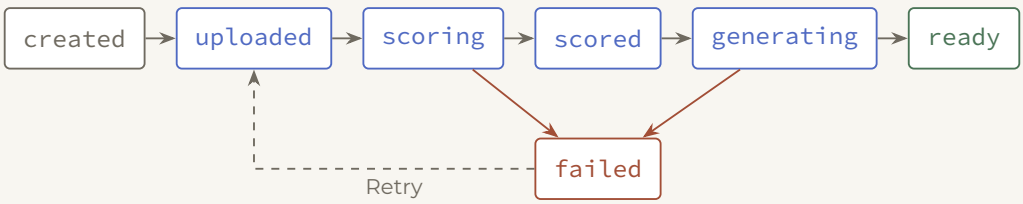


Figure 18: Lifecycle of a training session. *Retry* resumes from the last step that succeeded.

Status	Shown as	Meaning
`created`	Waiting for video	Session started, no video uploaded yet.
`uploaded`	Queued for analysis	Video received, waiting its turn.
`scoring`	Analysing video	The video is being analysed.
`scored`	Queued for report	Scores are ready; the report is waiting to be written.
`generating`	Writing report	The report is being written.
`ready`	Report ready	The report can be opened.
`failed`	Failed	Something went wrong; the page shows the reason and a **Retry** button.

05.2 Report and appeal status

Shown as	Meaning
Stamp *Pending review*	Your instructor has not completed a review of this report yet.
Stamp *Reviewed*	The review is complete; the instructor's evaluation is in section 05.
Appeal pending	Sent; waiting for the instructor. The student can still withdraw it.
Appeal accepted · +*n*	The finding is overturned and the score raised by *n* points.
Appeal rejected	The finding stands; the instructor's reasoning is shown.

05.3 Troubleshooting

I cannot sign in.
Check the email address and password (the password is case-sensitive). If it still fails, ask your course administrator to check that your account exists.

My video is rejected.
Only MP4, WebM, MOV and MKV files are accepted, up to 1 GB by default. Empty files are refused.

Processing failed.
Open the session (from *Recordings in progress* on the dashboard) and press **Retry**. If it fails again, record a new session with better lighting and framing.

Accept report is greyed out.
You have an appeal pending on that report. Wait for the decision or withdraw the appeal first.

There is no Appeal link on an event.
Only warnings and issues can be appealed, each once, up to five appeals per report. Normal steps are hidden until you press **Show all *n* events**.

The print dialog did not open after Export PDF.
Allow pop-ups for the site, or press **Save as PDF** at the top of the print page once the video stills have loaded.

An instructor cannot see a student.
Instructors only see students in the groups they teach.